Computer Graphics and Man-Machine Interface

Dr. Manar Donia

CMPn322 - CMPN322 - CMP 422

Course Objectives



The course discusses the fundamental concepts, aspects, and terminologies of computer graphics.



It presents the important drawing algorithm, polygon fitting, clipping, and 2D transformation curves and an introduction to 3D transformation.



It provides the basics of the OpenGL application programming interface that allows for developing programming skills in Computer Graphics.

Syllabus(CMPn322: Computer Graphics and Man-Machine Interface)

- **Basics of computer graphics**
  - **Computer generated Pictures and Raster Images.**
  - **Elements of computer-Generated Pictures**
  - **Drawing a Polylines and polygons.**
  - **Drawing a General functions**
- **Filling a region:**
  - **Rows based filling**
  - **Column based filling**
  - **Seed filling**
- **Transformations**
  - **2D transformations**
  - **3D transformations**
  - **Composite transformations**
  - **Inverse transforms**
- **Projection**
  - **Parallel Projection**
  - **Perspective Projection**
- **Lightening based on local reflection model.**
- **Containment and Clipping**

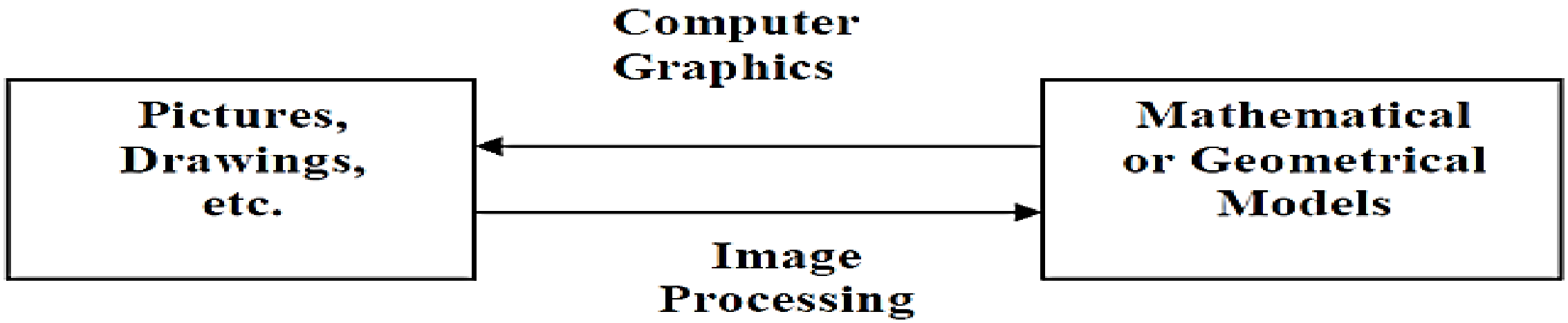
Assessment Method	Timing	Grade
Semester Work: quizzes assignments and reports	**Bi-Weekly**	**20**
Mid-Term Exam	**7th Week**	**20**
Practical Exam	**15th Week**	**20**
Written Exam	**14th Week**	**40**
Total		**100**

Course Assessment(2018-2020)

Assessment Method	Timing	Grade
Semester Work: quizzes assignments and reports	**Bi-Weekly**	**10**
Mid-Term Exam	**7th Week**	**10**
Practical Exam	**15th Week**	**20**
Written Exam	**14th Week**	**60**
Total		**100**

Course Assessment (2012)

Computer graphics and Image processing



computer graphics are concerned with the creation of images

Image Processing: analyze pictures to derive descriptions (often in mathematical or geometrical forms) of objects appeared in the pictures.

What is Computer graphics?

- Computer graphics means creation, storage and manipulation of models and images.



Computer graphics is a cross-disciplinary field in which **physics, mathematics, human perception, human-computer interaction, engineering, graphic design, and art all play important roles**.

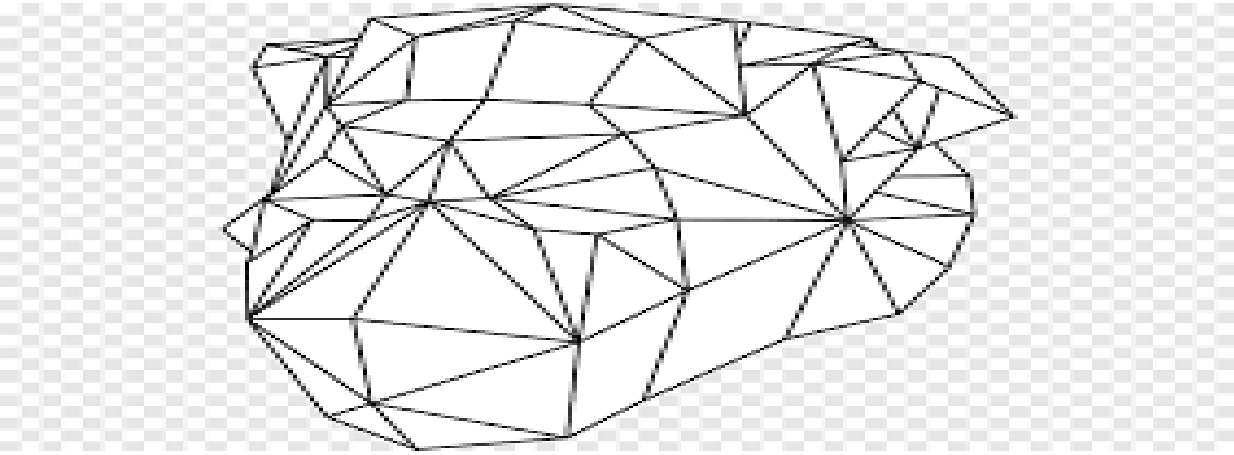
- We use **physics** to model light and to perform simulations for animation.
- We use **mathematics t**o describe shape and motion amount and direction.
- **Human perceptual abilities** determine our allocation of resources—we don't want to spend time rendering things that will not be noticed.
- We use **engineering** in optimizing the allocation of bandwidth, memory, and processor time. Graphic design and art combine with human-computer interaction to make the computer-to-human direction of communication most effective.



Computer graphics Model

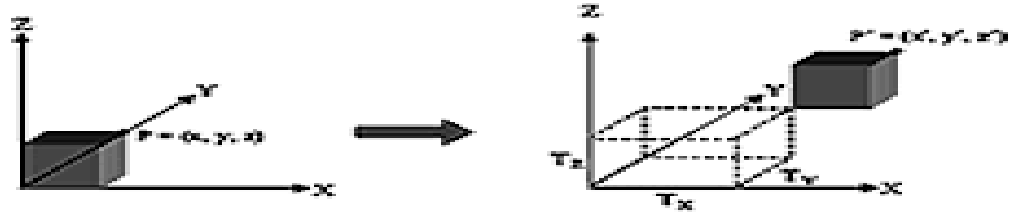
- Geometric model : describes how objects appears (it defines attributes that describe color texture or shape of the object)

- Mathematical model: is a model of a physical process in which it describes how object moves



Geometric model

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Mathematical model

Computer Graphics Applications

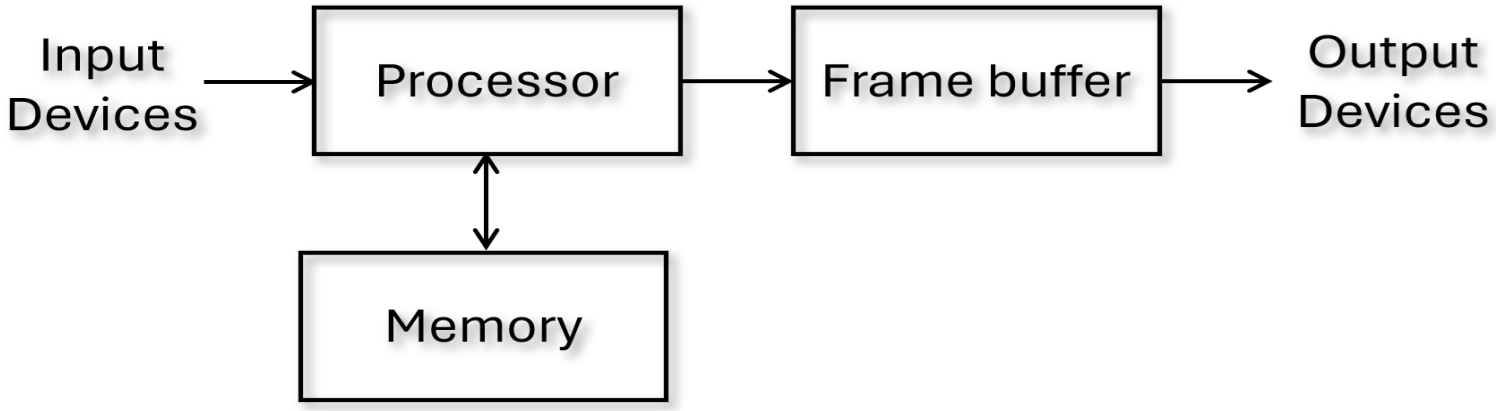
- Computer graphics has number of applications:



Computer Graphics system

- There are two major types of computer graphics systems:
  - Non-interactive computer graphics:
    - The picture is produced on the monitor, and the user does not have any controlled over the image as shown in animation movies.
    - It involves only **one-way communication** between the computer and the user where uses can see the produced image, and cannot make any change in the image
  - Interactive computer graphics:
    - The picture is produced on the monitor, and the user can make any change by editing modification with an input device
    - It involves **two-way communication** between the computer and the user as shown in games.

Computer graphic system



- I/P device are the h/w that is used to transfers I/P to the computer. The data can be in the form of text , graphics or sound
- Processor (CPU &GPU)
  - CPU gets the specifications for a geometric shape from input device and hands the geometric data with additional operations) to the GPU
  - The GPU rasterizes the data by deciding which pixels to use to represent the data then places the pixel data into the framebuffer to show on the display screen
- Frame buffer: It is a block of memory dedicated to graphics output which holds data representing all the pixels in a complete video frame.
- O/P device displays data from memory of the computer.

Computer graphic system

- Different types of data are retrieved from input devices called **input primitives.**
- Locator Devices (Interactive drawing)
  - Input Primitive: Coordinate position (x,y)



- String Devices(Text input)
  - Input Primitive: A string of characters



- Valuator (Games)
  - Input primitive: Scalar values (typically between 0 and 1)
  - can be used to fix the length of a line, the speed of an action, or perhaps the size of a picture



Computer graphic system

- Choice (Interactive menu selection)
  - Input primitive: A selection from a list of options.

- Pick (identify a portion of a picture for further processing)
  - Input primitive: Selection of a part of the screen.

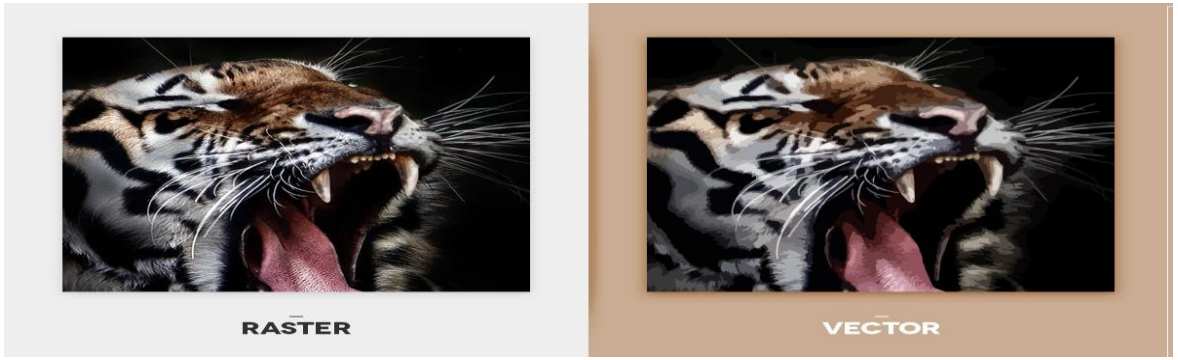


Computer graphic system

- Graphics display device:
  - **Vector displays** it construct pictures by drawing straight-line segments between specified endpoint positions.
    - Picture information is then stored as a set of line-drawing instructions.
    - It best for printing since it is composed of **a series of mathematical curves**. As a result, vector graphics print precisely even when they are enlarged
    - **It's known as scalable vector graphics (SVG)**
  - **Raster displays** uses a frame buffer to store the color value for each screen position (pixel).
    - The larger the image, the more disk space the image file will take up. As an example, a 200 x 200 image requires less space than 500 x 500 one
    - Scaling down these images is easy but enlarging a bitmap makes it pixelated or simply blurred.

Computer graphic system

- Graphics display device:
  - Raster
    - Composed of pixels
    - It produce large file size
    - It provide more details shape
    - Used for details Images such as photo
  - Vector
    - Composed of connected sequence of points (paths)
    - It produce small file size
    - It provide smooth shape but limited to set of colors
    - Used for less details Images such as logos

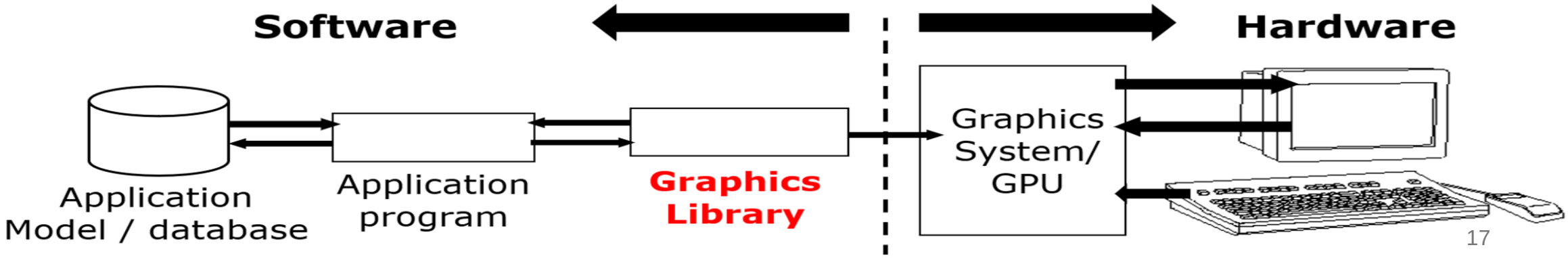




If your project requires scalable shapes and solid colors, vector is the best choice as for logos and icons
if your project requires complex color blends, raster is the preferred format as for digital photos.

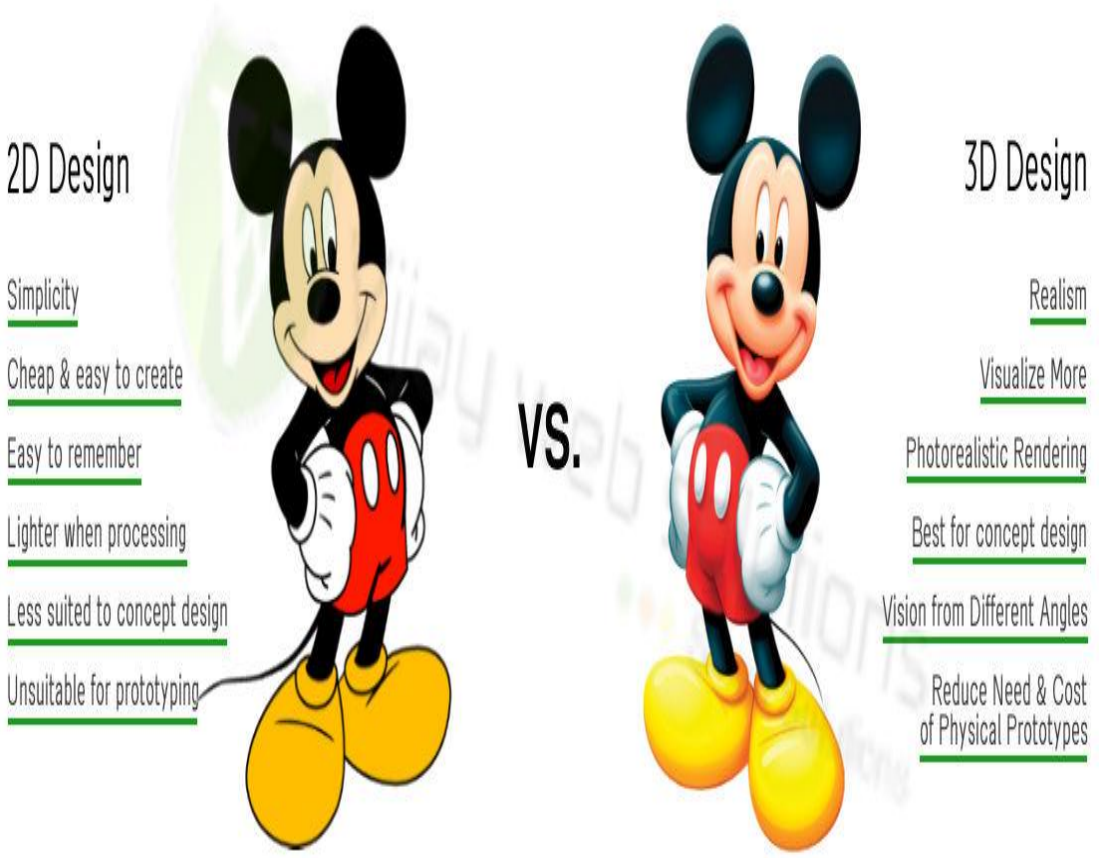
Conceptual Framework of Computer Graphics

- Interactive computer graphics consists of two major components at two levels:
  - Hardware level: the computer receives input from interactive devices and show images on display devices.
  - Software level: it involves three components
    - Application program→ creates, store into and retrieved from application model such as word, excel, paint, or photoshop program (handles user's input)
    - Application model→ it capture all the data or objects to be pictured on the screen, that stored in application Database.
    - Graphics library or application programmer's interface (API)→ is a set of procedures for handling graphics Such as OpenGL, Direct3D or Java3D. it is intermediary between application and display hardware (Graphics System)



2D or 3D first

- 2D is a good steppingstone towards 3D many issues much easier to understand in 2D
  - you can't be a 3D animator unless you have learned some 2D first.
  - No need to simulate lights, cameras, etc.
  - Intro to modelling (SketchUp) vs rendering (use colors, materials, and textures to create a realistic image of the model).
- 2D is still important and the most common use of computer graphics as in advertising and many other fields.



Graphic Library

OpenGl stands for Open Graphics Library producing 2D and 3D graphics..

OpenGL supports **every major operating system** (Windows, Linux, Android and Mac OS).

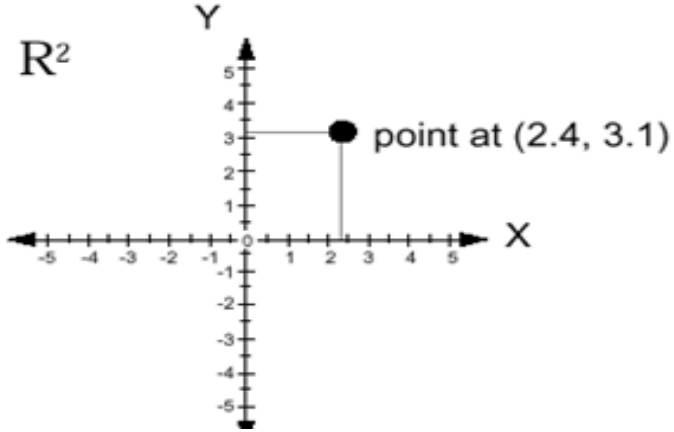
OpenGl supports a wide variety of programming languages such as C++, Python, Java, etc..

But Primary interface of OpenGl is in C++

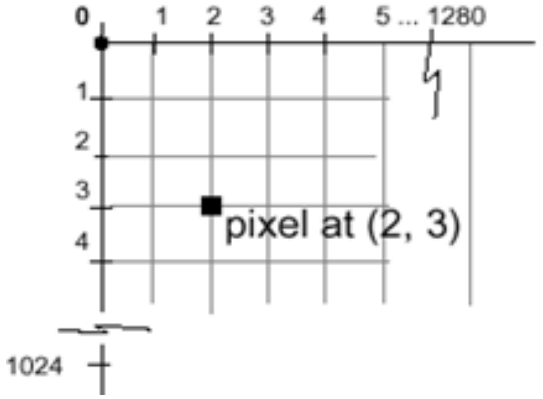
Coordinate Systems

When you draw lines or plot points on a simple flat computer screen, you need to specify its position in terms of its coordinates(x, y).

- Cartesian coordinates in math, engineering
  - Modeled as floating point
  - X increasing right, Y increasing up

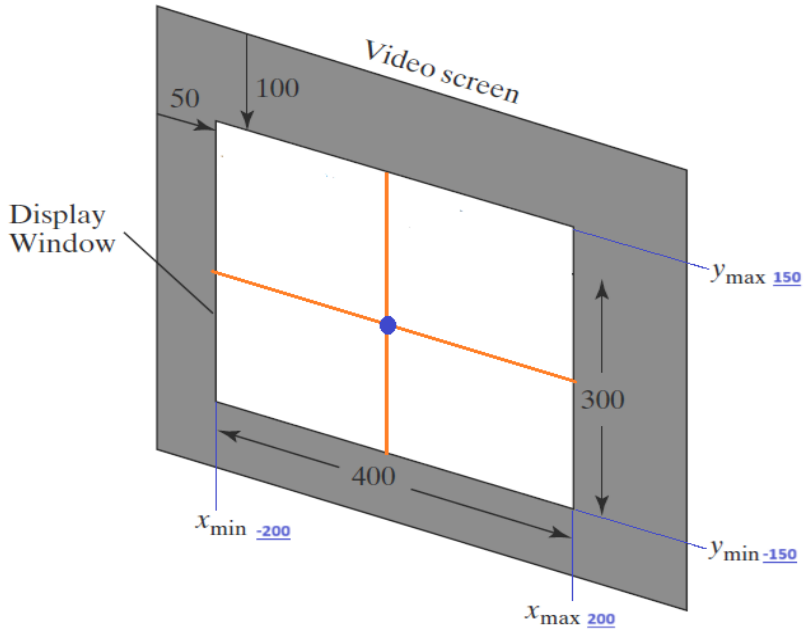


- Display (physical) coordinates
  - Integer values only
  - X increasing right, Y increasing down by 1 unit = 1 pixel

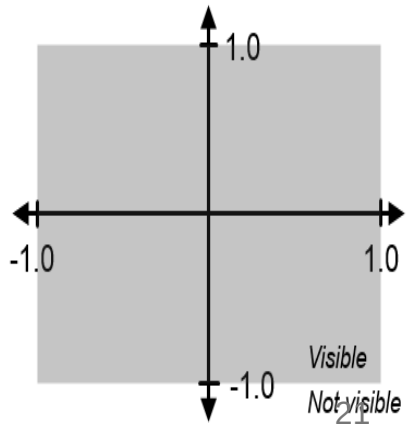


Coordinate Systems

- In OpenGL when you create a window to draw in, we use glut library which is the basic window interface, the window is measured physically in terms of pixels.
  - For example,
    - We specify a display window with a width of 400 pixels and a height of 300 pixels using **glutInitWindowSize (400, 300);**
    - We use the **glutInitWindowPosition** function to give an initial location for the upper left corner of the display window using **glutInitWindowPosition (50, 100);**
    - Then specify the region of Cartesian space that occupies the window using **gluOrtho2D (xmin, xmax, ymin, ymax);**
      - This region is known as the clipping region which is the minimum and maximum x and y values that are inside the window.

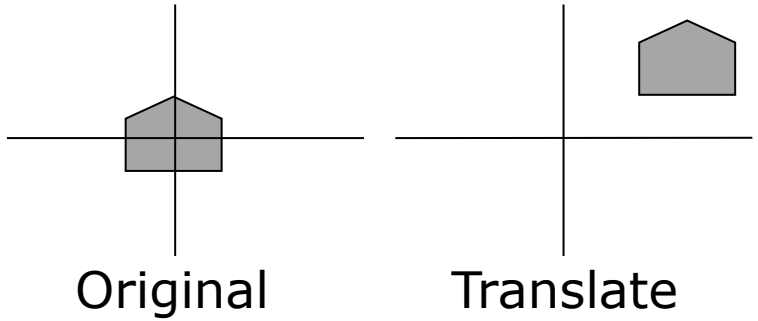


OpenGL use normalized device coordinates after each vertex That is, the coordinates of each vertex should be between -1.0 and 1.0;

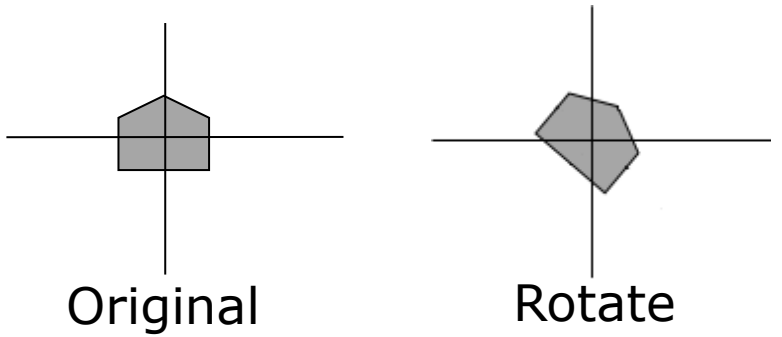


Transformations

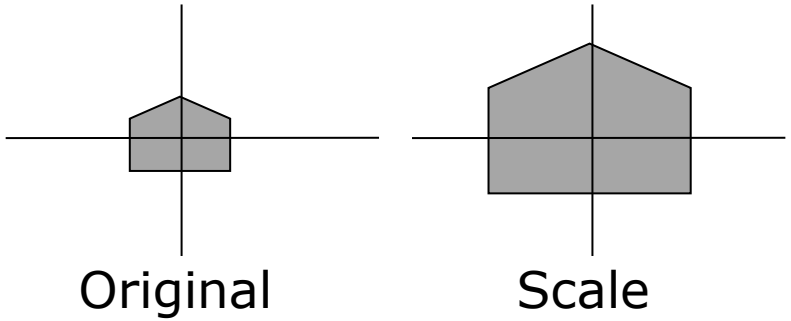
- Geometric Transformations in 2D (note the z-coordinate is 0)




```
model *= glm::translate(.1, .1, 0);
```




```
model *= glm::rotate(-45, glm::vec3(0, 0, 1));
```

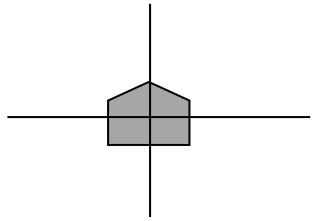



```
model *= glm::scale(2, 2, 1);
```

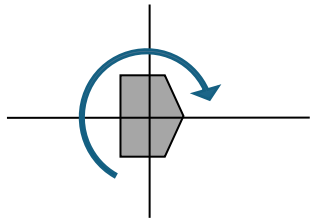
- Positive angles rotate counter-clockwise, here about the origin (i.e., Z-axis as vector)

Transformations

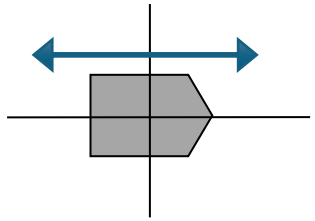
- Transformations can be **composed** (combined by multiplying matrices) but are NOT commutative, so proper order is vital

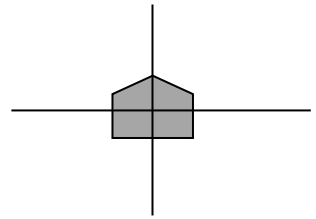


```
model *= glm::rotate(-90, glm::vec3(0, 0, 1));
```

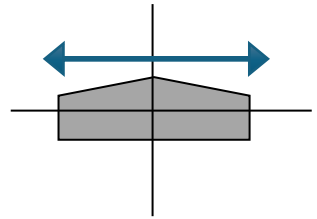


```
model *= glm::scale(2, 1, 1);
```

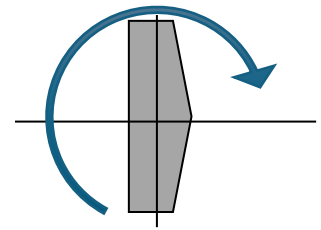




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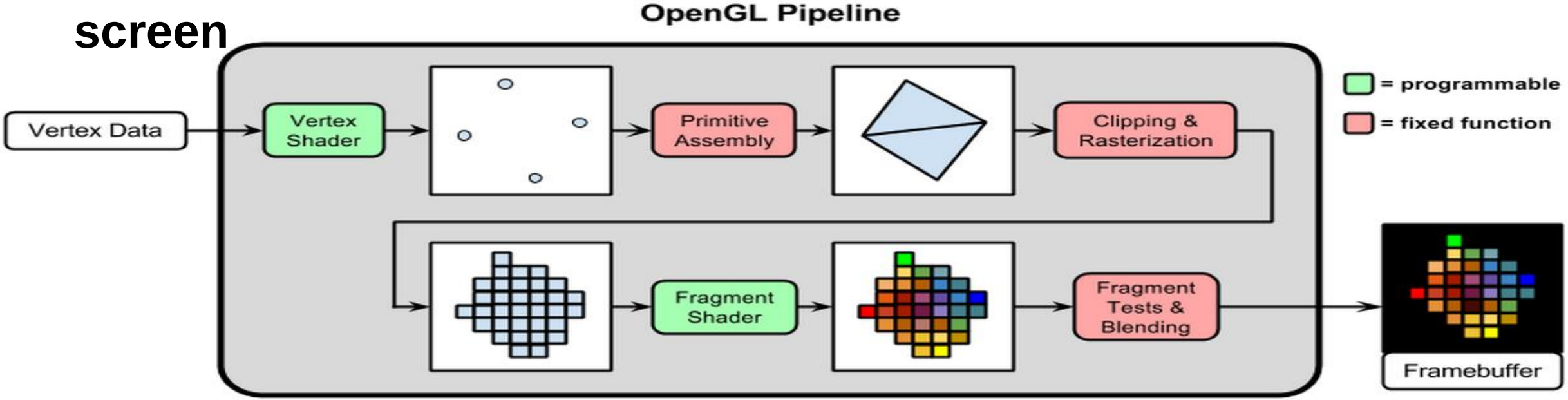


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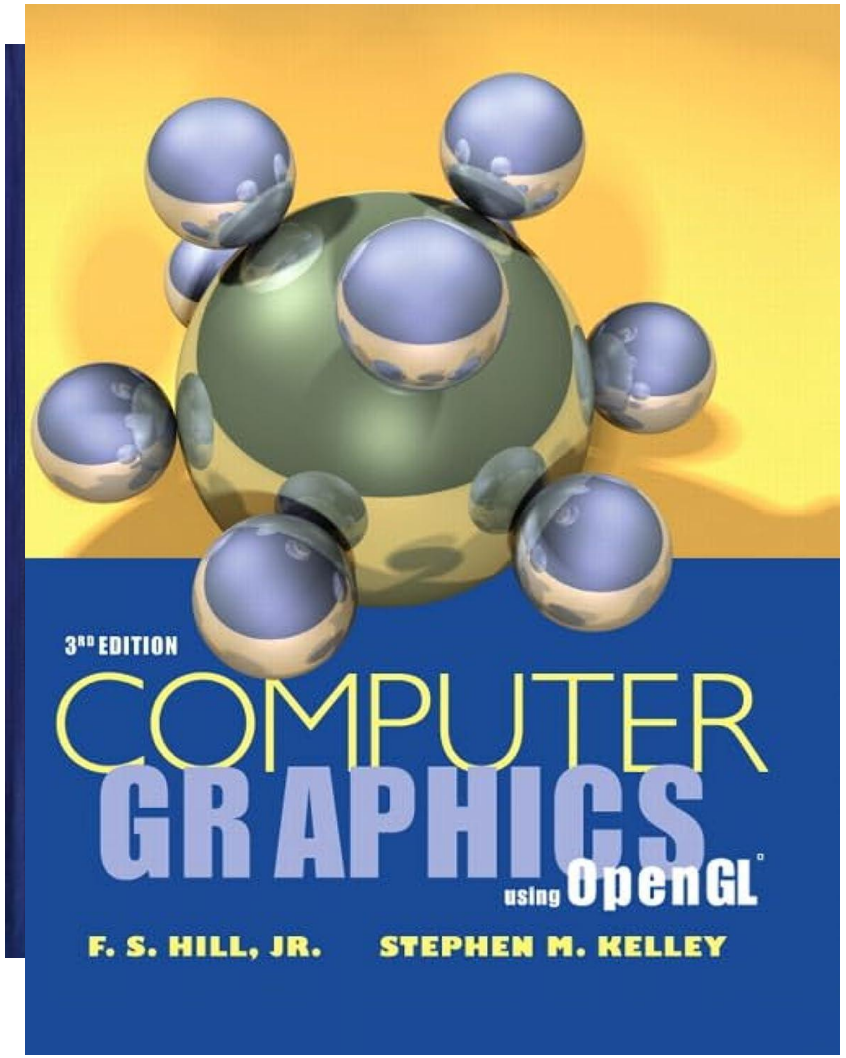
OpenGL pipeline

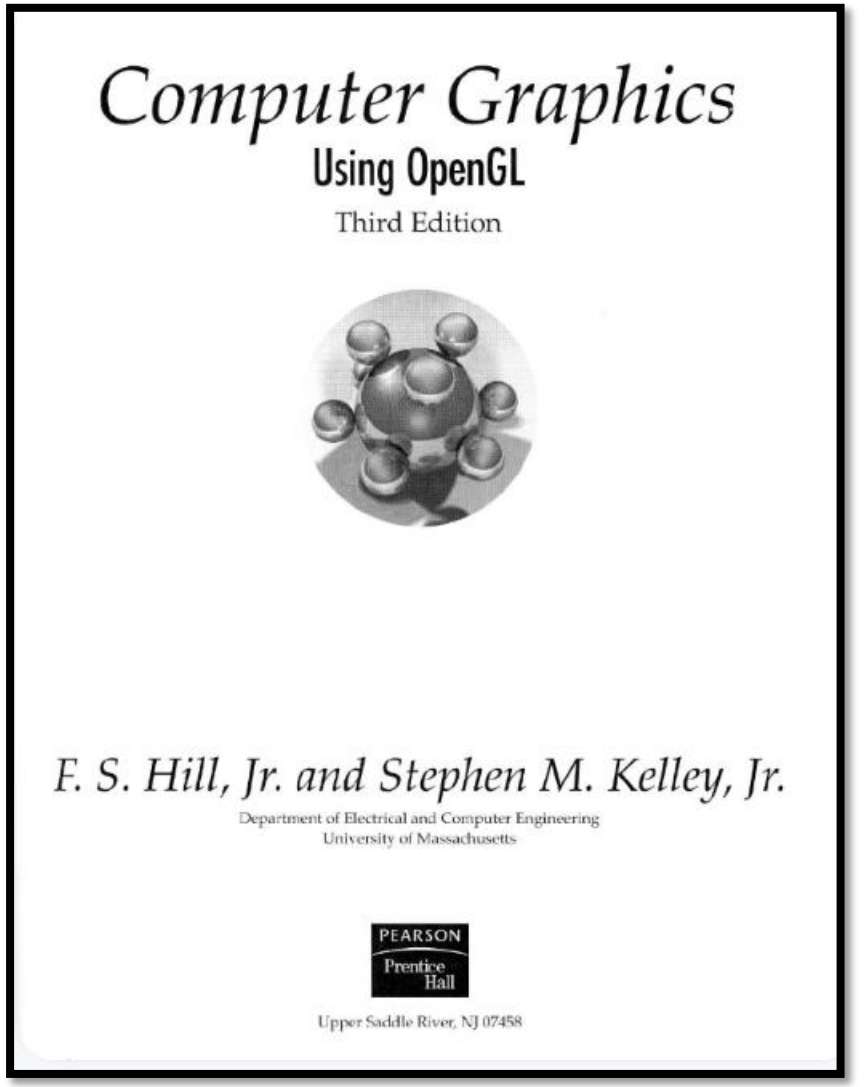
Graphics pipeline is a conceptual model that describes what steps a graphics system needs to perform to **render a 3D scene to a 2D screen**



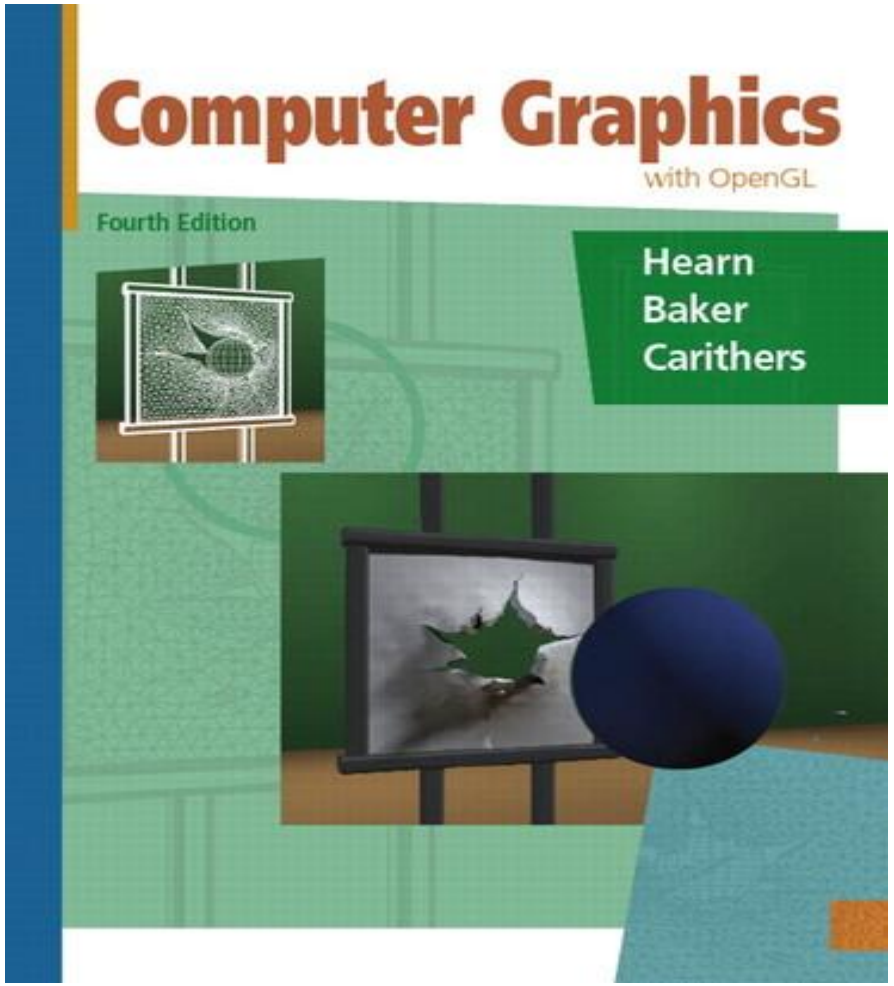
The vertex shader → define object vertices to move object around(translation ,scaling, rotation)
Clipping → cutting vertices that will not on the screen
Rasterization→ creating pixels for each part of the object
Fragment shader→ add values for each pixel (coloring object shape)

Textbook


3RD EDITION
COMPUTER
GRAPHICS
using OpenGL
F. S. HILL, JR.
STEPHEN M. KELLEY


Computer Graphics
Using OpenGL
Third Edition
F. S. Hill, Jr. and Stephen M. Kelley, Jr.
Department of Electrical and Computer Engineering
University of Massachusetts
PEARSON
Prentice Hall
Upper Saddle River, NJ 07458

Textbook



Computer Graphics with Open GL, 4th Edition

Donald D. Hearn, University of Illinois at Urbana-Champaign

M. Pauline Baker, Indiana University-Purdue University Indianapolis

Warren Carithers, Rochester Institute of Technology

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Thank You